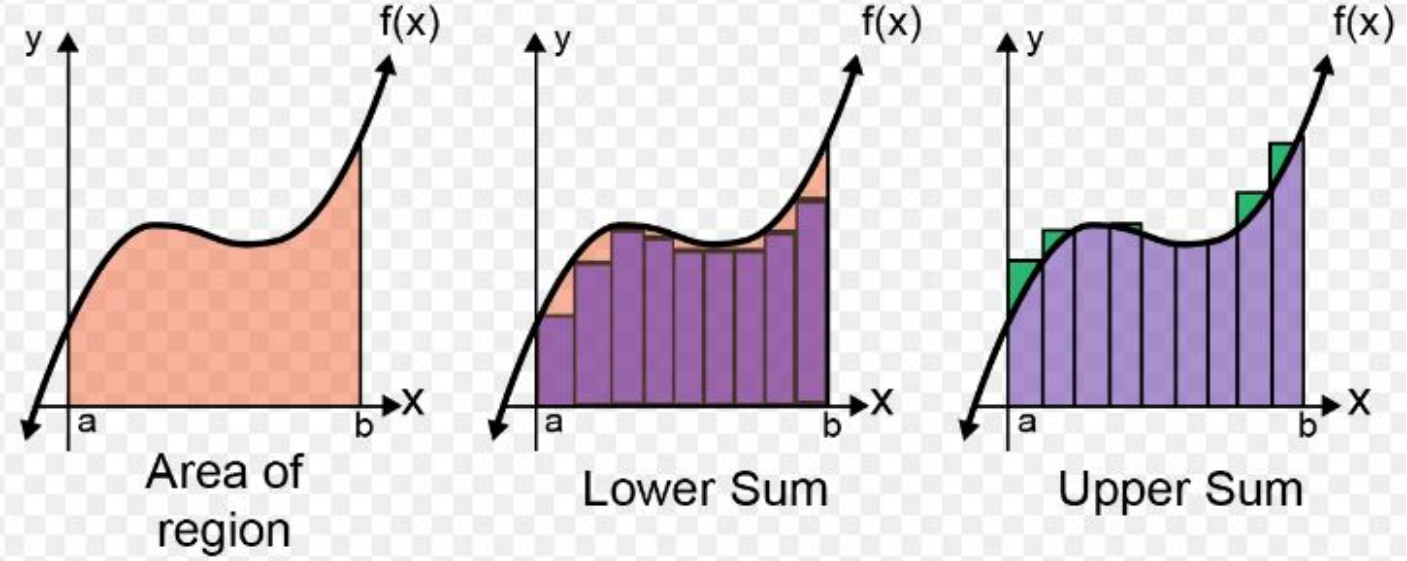
MAT 221 - CALCULUS I
Lesson 12 - Riemann Sums

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Riemann Sums



- A Riemann sum is a method used to approximate the area under a curve by dividing the area into rectangles and summing their areas.
- Let f be a function defined on a closed interval $[a, b]$, of real numbers, and a regular partition $(x_0, x_1, ..., x_n)$ of $[a, b]$, i.e $a = x_0 < x_1 < ... < x_n = b$.
- A Riemann sum of f over $[a, b]$ with a partition $(x_0, x_1, ..., x_n)$ is defined as;

$$f(x_1^*)\Delta x_1 + f(x_2^*)\Delta x_2 + ... + f(x_n^*)\Delta x_n = \Sigma_{i=1}^{n} f(x_i^*)\Delta x_i. \quad (1)$$

where $\Delta x_i = \frac{b-a}{n}$ and $x_i^* = a + i\Delta x_i$.

Definite Integral using Regular Partitions

Let f be continuous on $[a, b]$ with $a < b$. Then

$$\int_a^b f(x)dx = \lim_{\Delta x \to 0} \sum_{i=1}^{n} f(x_i^*)\Delta x = \lim_{n \to \infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

Remark

For a regular partition, where $\Delta x_i \to 0$, means $n \to \infty$, where n is the number of subintervals.

Formulas for Sums

$$(i) \sum_{i=1}^{\infty} c = nc, (ii) \sum_{i=1}^{\infty} i = \frac{n(n+1)}{2}, (iii) \sum_{i=1}^{\infty} i^2 = \frac{n(n+1)(2n+1)}{6},$$

$$(iv) \sum_{i=1}^{\infty} i^3 = \frac{n^2(n+1)^2}{4}.$$

Example 1

Use the limit and Riemann sums to evaluate the integral

$$\int_1^2 (2x^2 - 1)dx.$$

Solution

$a = 1, b = 2, \Delta x = \frac{2-1}{n} = \frac{1}{n}$ while $x_i^* = a + i\Delta x = 1 + \frac{1}{n}i$.
$f(x_i^*) = 2(1 + \frac{1}{n}i)^2 - 1 = 2(1 + \frac{2}{n}i + \frac{1}{n^2}i^2) - 1$.
$f(x_i^*)\Delta x = \frac{1}{n} + \frac{4}{n^2}i + \frac{2}{n^3}i^2$.
Riemann sum:

$$\sum_{i=1}^{n} f(x_i^*)\Delta x = \sum_{i=1}^{n} \frac{1}{n} + \frac{4}{n^2}\sum_{i=1}^{n} i + \frac{2}{n^3}\sum_{i=1}^{n} i^2.$$

$$= \frac{1}{n}.n + \frac{4}{n^2}\left[\frac{n(n+1)}{2}\right] + \frac{2}{n^3}\left[\frac{n(n+1)(2n+1)}{6}\right]$$

$$= 1 + 2\left(1 + \frac{1}{n}\right) + \frac{1}{3}\left(2 + \frac{3}{n} + \frac{1}{n^3}\right)$$

$$\int_1^2 (2x^2-1)dx = \lim_{n\to\infty} \sum f(x_i^*)\Delta x = \lim_{n\to\infty} \left[1 + 2\left(1 + \frac{1}{n}\right) + \frac{1}{3}\left(2 + \frac{3}{n} + \frac{1}{n^3}\right)\right.$$

$$= 1 + 2 + \frac{2}{3} = 3\frac{2}{3}$$

Example 2

Use the limit definition to evaluate the integral

$$\int_0^2 (x^3 - 9)dx.$$

Solution $a = 0, b = 1$, $\Delta x = \frac{2-0}{n} = \frac{2}{n}$ while $x_i^* = a + i\Delta x = \frac{2}{n}i$.
$f(x_i^*) = \left(\frac{2}{n}i\right)^3 - 9 = \frac{8}{n^3}i^3 - 9$.
$f(x_i^*)\Delta x = \frac{2}{n}\left(\frac{8}{n^3}i^3 - 9\right) = \frac{16}{n^4}i^3 - \frac{18}{n}$ Riemann sum:

$$\sum_{i=1}^{n} f(x_i^*)\Delta x = \sum_{i=1}^{n}\left(\frac{16}{n^4}i^3 - \frac{18}{n}\right) = \frac{16}{n^4}\sum_{i=1}^{n} i^3 - \sum_{i=1}^{n}\frac{18}{n}$$

$$= \frac{16}{n^4}.\frac{n^2(n+1)^2}{4} - \frac{18}{n}.n = 4\left(1 + \frac{2}{n} + \frac{1}{n^2}\right) - 18$$

$$\int_0^2 (x^3 - 9)dx = \lim_{n\to\infty}\sum f(x_i^*)\Delta x = \lim_{n\to\infty}\left[4\left(1 + \frac{2}{n} + \frac{1}{n^2}\right) - 18\right]$$

$$= 4 - 18 = -14.$$

Example 3

Use the limit definition to evaluate the integral

$$\int_0^1 (2 - x + 2x^2)dx.$$

Solution $a = 0, b = 1$, $\Delta x = \frac{1-0}{n} = \frac{1}{n}$ while $x_i^* = a + i\Delta x = \frac{1}{n}i$.
$f(x_i^*) = 2 - \frac{1}{n}i + \frac{2}{n^2}i^2$. Riemann sum:

$$\sum_{i=1}^{n} f(x_i^*)\Delta x = \sum_{i=1}^{n}\left(\frac{2}{n} - \frac{i}{n^2} + \frac{2}{n^3}i^2\right) = \frac{2}{n}.n - \frac{1}{n^2}\sum_{i=1}^{n} i + \frac{2}{n^3}\sum_{i=1}^{n} i^2.$$

$$= 2 - \frac{1}{2}\left(1 + \frac{1}{n}\right) + \frac{1}{3}\left(1 + \frac{1}{n}\right)\left(2 + \frac{1}{n}\right).$$

$$\int_0^1 (2-x+2x^2)dx = \lim_{n\to\infty}\sum f(x_i^*)\Delta x = \lim_{n\to\infty}\left[2 - \frac{1}{2}\left(1 + \frac{1}{n}\right) + \frac{1}{3}\left(1 + \frac{1}{n}\right)\left(2\right.\right.$$

$$= 2 - \frac{1}{2} + \frac{2}{3} = \frac{13}{6}.$$